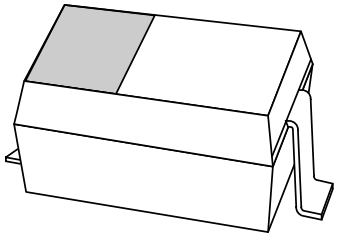
DATA SHEET



BZX399 series
Voltage regulator diodes

Product specification

1999 Jun 04

Philips
Semiconductors

PHILIPS

Voltage regulator diodes BZX399 series

FEATURES

- Total power dissipation: max. 300 mW
- Tolerance: ±5%
- Working voltage range: nom. 1.8 to 43 V (E24 range)
- Improved I_Z/V_Z characteristic at low currents (I_Z = 50 μA). This results in a noise free and sharp breakdown knee.

APPLICATIONS

- General regulation functions, where low noise at low currents is required
- Low power consumption applications (e.g. hand-held applications).

DESCRIPTION

Low-power low noise voltage regulator diodes in SOD323 plastic SMD package.

The diodes are available in the normalized E24 ±5% tolerance range. The series consists of 34 types with nominal working voltages from 1.8 to 43 V.

PINNING

PIN	DESCRIPTION
1	cathode
2	anode

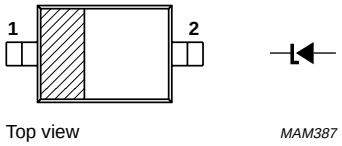


Fig.1 Simplified outline (SOD323) and symbol.

MARKING

TYPE NUMBER	MARKING CODE	TYPE NUMBER	MARKING CODE	TYPE NUMBER	MARKING CODE	TYPE NUMBER	MARKING CODE
BZX399-C1V8	B1	BZX399-C4V3	B0	BZX399-C10	BJ	BZX399-C24	BT
BZX399-C2V0	B2	BZX399-C4V7	BA	BZX399-C11	BK	BZX399-C27	BU
BZX399-C2V2	B3	BZX399-C5V1	BB	BZX399-C12	BL	BZX399-C30	BV
BZX399-C2V4	B4	BZX399-C5V6	BC	BZX399-C13	BM	BZX399-C33	BW
BZX399-C2V7	B5	BZX399-C6V2	BD	BZX399-C15	BN	BZX399-C36	BX
BZX399-C3V0	B6	BZX399-C6V8	BE	BZX399-C16	BP	BZX399-C39	BY
BZX399-C3V3	B7	BZX399-C7V5	BF	BZX399-C18	BQ	BZX399-C43	BZ
BZX399-C3V6	B8	BZX399-C8V2	BG	BZX399-C20	BR		
BZX399-C3V9	B9	BZX399-C9V1	BH	BZX399-C22	BS		

LIMITING VALUES

In accordance with the Absolute Maximum Rating System (IEC 134).

SYMBOL	PARAMETER	CONDITIONS	MIN.	MAX.	UNIT
I_F	continuous forward current		–	250	mA
I_{ZSM}	non-repetitive peak reverse current	t_p = 100 µs; square wave; T_{amb} = 25 °C prior to surge	see Tables 1 and 2		
P_{tot}	total power dissipation	T_{amb} = 25 °C; note 1	–	300	mW
T_{stg}	storage temperature		–65	+150	°C
T_j	junction temperature		–	150	°C

Note

1. Device mounted on a FR4 printed circuit-board.

ELECTRICAL CHARACTERISTICS

Total BZX399-C series

T_j = 25 °C unless otherwise specified.

SYMBOL	PARAMETER	CONDITIONS	MAX.	UNIT
V_F	forward voltage	I_F = 10 mA; see Fig.5	0.9	V
		I_F = 100 mA; see Fig.5	1.0	V
I_R	reverse current			
	BZX399-C1V8	V_R = 1 V	2	µA
	BZX399-C2V0	V_R = 1 V	1	µA
	BZX399-C2V2	V_R = 1 V	0.5	µA
	BZX399-C2V4	V_R = 1 V	0.2	µA
	BZX399-C2V7	V_R = 1 V	0.05	µA
	BZX399-C3V0	V_R = 1 V	0.02	µA
	BZX399-C3V3	V_R = 2 V	2	µA
	BZX399-C3V6	V_R = 2 V	1	µA
	BZX399-C3V9	V_R = 2 V	0.5	µA
	BZX399-C4V3	V_R = 2 V	0.1	µA
	BZX399-C4V7	V_R = 3 V	2	µA
	BZX399-C5V1	V_R = 3 V	1	µA
	BZX399-C5V6	V_R = 4 V	1	µA
	BZX399-C6V2	V_R = 5 V	0.1	µA
	BZX399-C6V8	V_R = 5 V	0.01	µA
	BZX399-C7V5	V_R = 5 V	0.1	µA
	BZX399-C8V2	V_R = 6 V	0.2	µA
	BZX399-C9V1	V_R = 7 V	0.1	µA
	BZX399-C10	V_R = 7 V	0.1	µA
	BZX399-C11	V_R = 8 V	0.05	µA
	BZX399-C12	V_R = 9 V	0.05	µA
	BZX399-C13	V_R = 10 V	0.05	µA
	BZX399-C15 to 43	$V_R = 0.7V_{Znom}$	0.01	µA

Table 1 Per type BZX399-**C1V8** to **C15**

T_j = 25 °C unless otherwise specified.

BZX399-C XXX	WORKING VOLTAGE V_Z (V) I_Z = 50 μA		VOLTAGE CHANGE ΔV_Z(V) (note 1)	TEMP. COEFF. S_Z (mV/K) I_Z = 50 μA (see Figs 2, 3 and 4)	DIODE CAP. C_d (pF) at f = 1 MHz; V_R = 0 V	NON-REPETITIVE PEAK REVERSE CURRENT I_{ZSM} (A) at t_p = 100 μs; T_{amb} = 25 °C
	Tol. ±5%					
	MIN.	MAX.	MAX.	TYP.	MAX.	MAX.
1V8	1.71	1.89	0.65	–0.85	425	6.0
2V0	1.90	2.10	0.70	–0.95	410	6.0
2V2	2.09	2.31	0.75	–1.05	390	6.0
2V4	2.28	2.52	0.80	–1.15	370	6.0
2V7	2.57	2.84	0.85	–1.35	350	6.0
3V0	2.85	3.15	0.90	–1.50	325	6.0
3V3	3.14	3.47	0.93	–1.65	310	6.0
3V6	3.42	3.78	0.95	–1.80	300	6.0
3V9	3.71	4.10	0.97	–1.95	290	6.0
4V3	4.09	4.52	0.99	–2.05	280	6.0
4V7	4.47	4.94	0.97	–1.90	275	6.0
5V1	4.85	5.36	0.60	0.15	300	5.0
5V6	5.32	5.88	0.20	1.75	275	4.0
6V2	5.89	6.51	0.10	2.35	250	3.0
6V8	6.46	7.14	0.10	3.00	215	3.0
7V5	7.13	7.88	0.15	3.60	170	3.0
8V2	7.79	8.61	0.15	4.25	150	3.0
9V1	8.65	9.56	0.10	5.00	120	3.0
10	9.50	10.50	0.10	5.80	110	3.0
11	10.45	11.55	0.11	6.70	110	2.5
12	11.40	12.60	0.12	7.65	105	2.5
13	12.35	13.65	0.13	8.60	105	2.5
15	14.25	15.75	0.15	10.50	100	2.0

Note

1. ΔV_Z = V_Z at 100 μA minus V_Z at 10 μA.

Table 2 Per type BZX399-**C16** to **C43**

T_j = 25 °C unless otherwise specified.

BZX399- C XXX	WORKING VOLTAGE V_Z (V) I_Z = 50 μA		VOLTAGE CHANGE ΔV_Z(V) (note 1)	TEMP. COEFF. S_Z (mV/K) I_Z = 50 μA (see Fig.4)	DIODE CAP. C_d (pF) at f = 1 MHz; V_R = 0 V	NON-REPETITIVE PEAK REVERSE CURRENT I_{ZSM} (A) at t_p = 100 μs; T_{amb} = 25 °C
	Tol. ±5%					
	MIN.	MAX.	MAX.	TYP.	MAX.	MAX.
16	15.20	16.80	0.16	11.4	95	1.5
18	17.10	18.90	0.18	13.3	95	1.5
20	19.00	21.00	0.20	15.3	90	1.5
22	20.90	23.10	0.22	17.2	85	1.25
24	22.80	25.20	0.24	19.2	80	1.25
27	25.65	28.35	0.27	22.0	75	1.0
30	28.50	31.50	0.30	25.2	65	1.0
33	31.35	34.65	0.30	28.5	60	0.9
36	34.20	37.80	0.30	32.0	60	0.8
39	37.05	40.95	0.30	35.1	60	0.7
43	40.85	45.15	0.30	39.4	55	0.6

Note

1. ΔV_Z = V_Z at 100 μA minus V_Z at 10 μA.

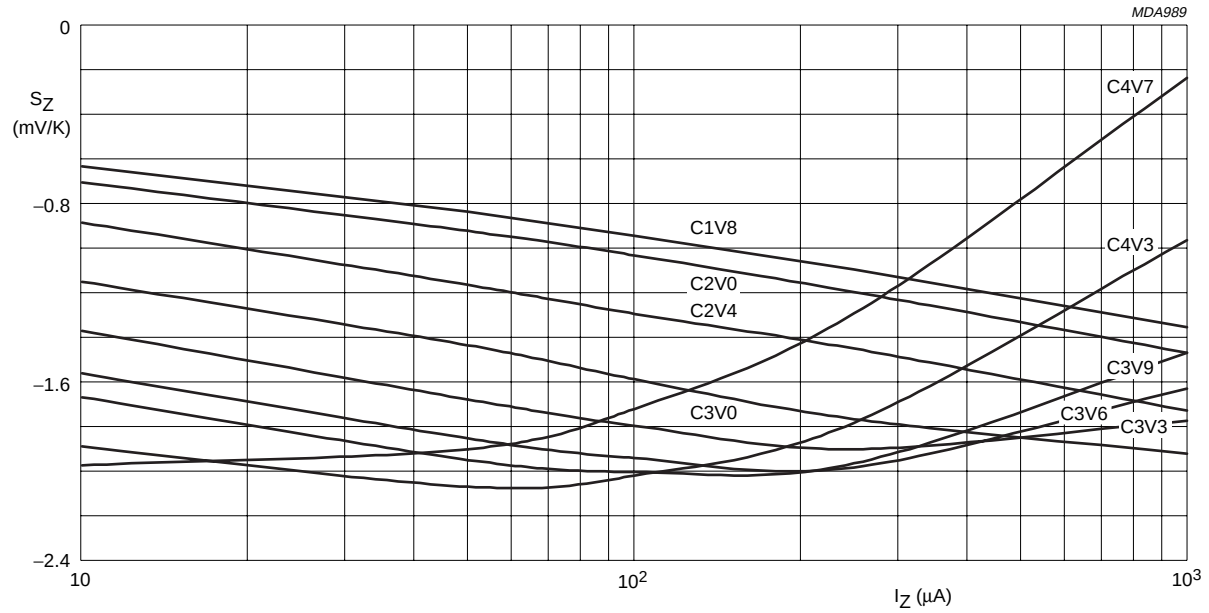
THERMAL CHARACTERISTICS

SYMBOL	PARAMETER	CONDITIONS	VALUE	UNIT
$R_{th\ j\text{-}a}$	thermal resistance from junction to ambient	note 1	415	K/W
$R_{th\ j\text{-}s}$	thermal resistance from junction to soldering point	note 2	110	K/W

Notes

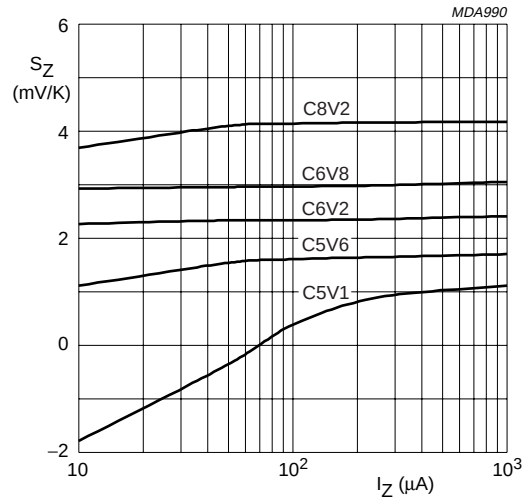
1. Device mounted on a FR4 printed circuit-board.
2. Soldering point of the cathode tab.

GRAPHICAL DATA



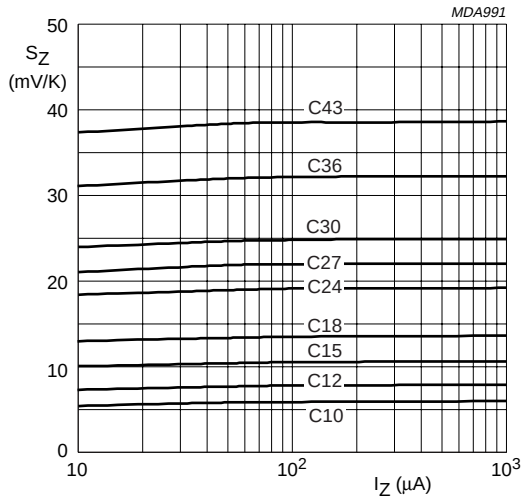
T_j = 25 to 150 °C.

Fig.2 Temperature coefficient as a function of working current; typical values.



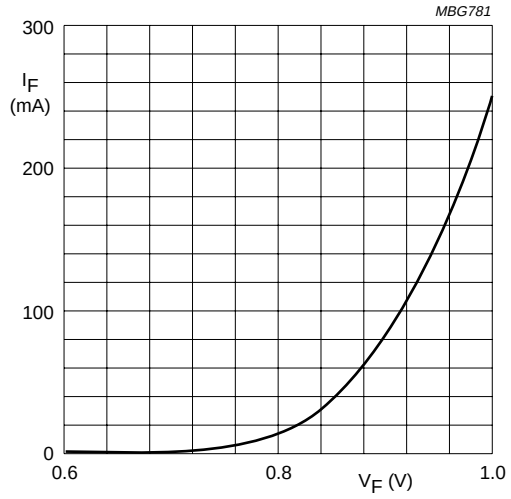
T_j = 25 to 150 °C.

Fig.3 Temperature coefficient as a function of working current; typical values.



T_j = 25 to 150 °C.

Fig.4 Temperature coefficient as a function of working current; typical values.

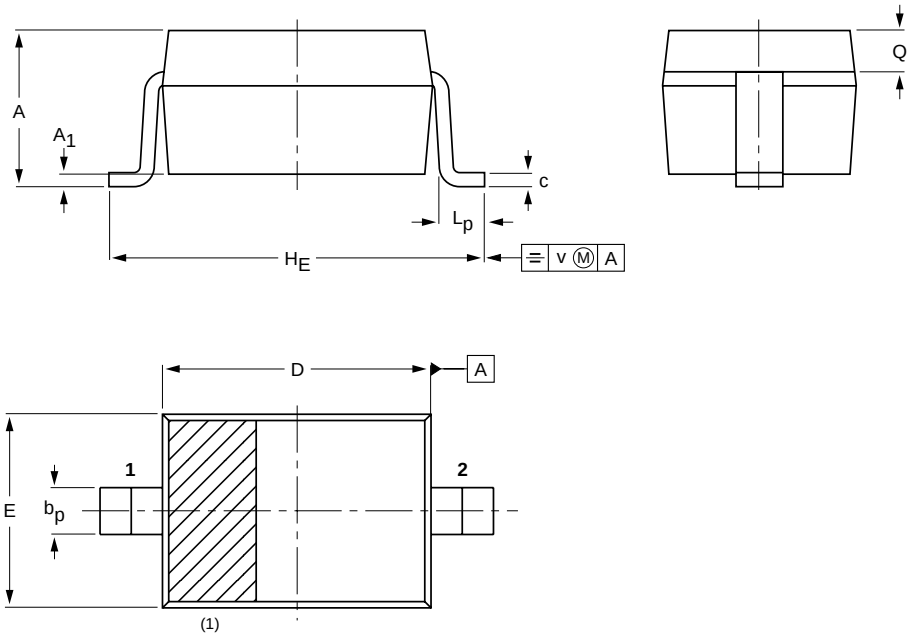


T_j = 25 °C.

Fig.5 Forward current as a function of forward voltage; typical values.

PACKAGE OUTLINE

Plastic surface mounted package; 2 leads **SOD323**



DIMENSIONS (mm are the original dimensions)

UNIT	A	A_1 max.	b_p	c	D	E	H_E	L_p	Q	v
mm	1.1 0.8	+ 0.05 – 0.05	0.40 0.25	0.25 0.10	1.8 1.6	1.35 1.15	2.7 2.3	0.45 0.15	0.25 0.15	0.2

Note

1. The marking bar indicates the cathode.

OUTLINE VERSION	REFERENCES				EUROPEAN PROJECTION	ISSUE DATE
	IEC	JEDEC	EIAJ			
SOD323						98-09-14

DEFINITIONS

Data sheet status	
Objective specification	This data sheet contains target or goal specifications for product development.
Preliminary specification	This data sheet contains preliminary data; supplementary data may be published later.
Product specification	This data sheet contains final product specifications.
Limiting values	
Limiting values given are in accordance with the Absolute Maximum Rating System (IEC 134). Stress above one or more of the limiting values may cause permanent damage to the device. These are stress ratings only and operation of the device at these or at any other conditions above those given in the Characteristics sections of the specification is not implied. Exposure to limiting values for extended periods may affect device reliability.	
Application information	
Where application information is given, it is advisory and does not form part of the specification.	

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These products are not designed for use in life support appliances, devices, or systems where malfunction of these products can reasonably be expected to result in personal injury. Philips customers using or selling these products for use in such applications do so at their own risk and agree to fully indemnify Philips for any damages resulting from such improper use or sale.

NOTES

NOTES

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